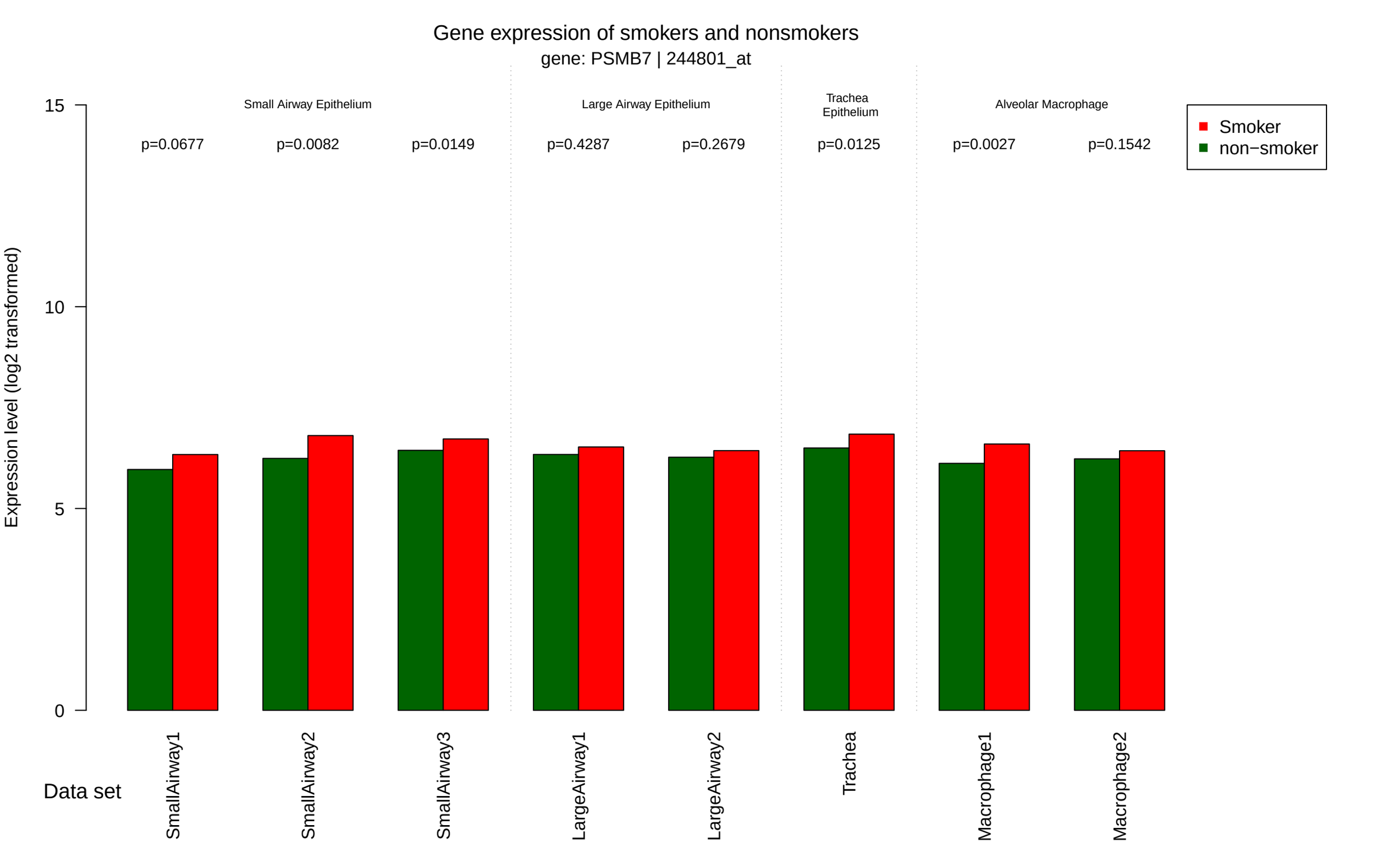
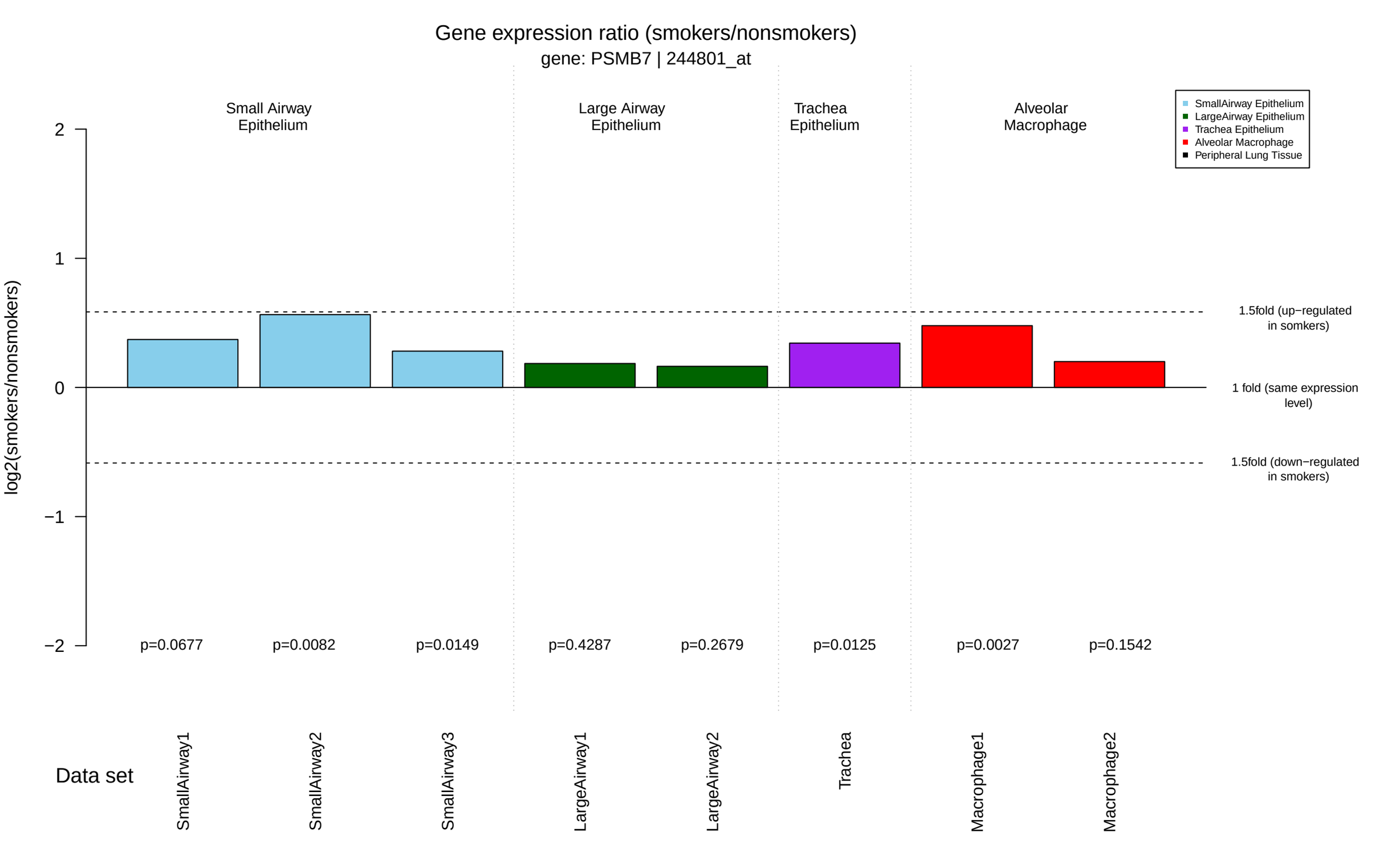
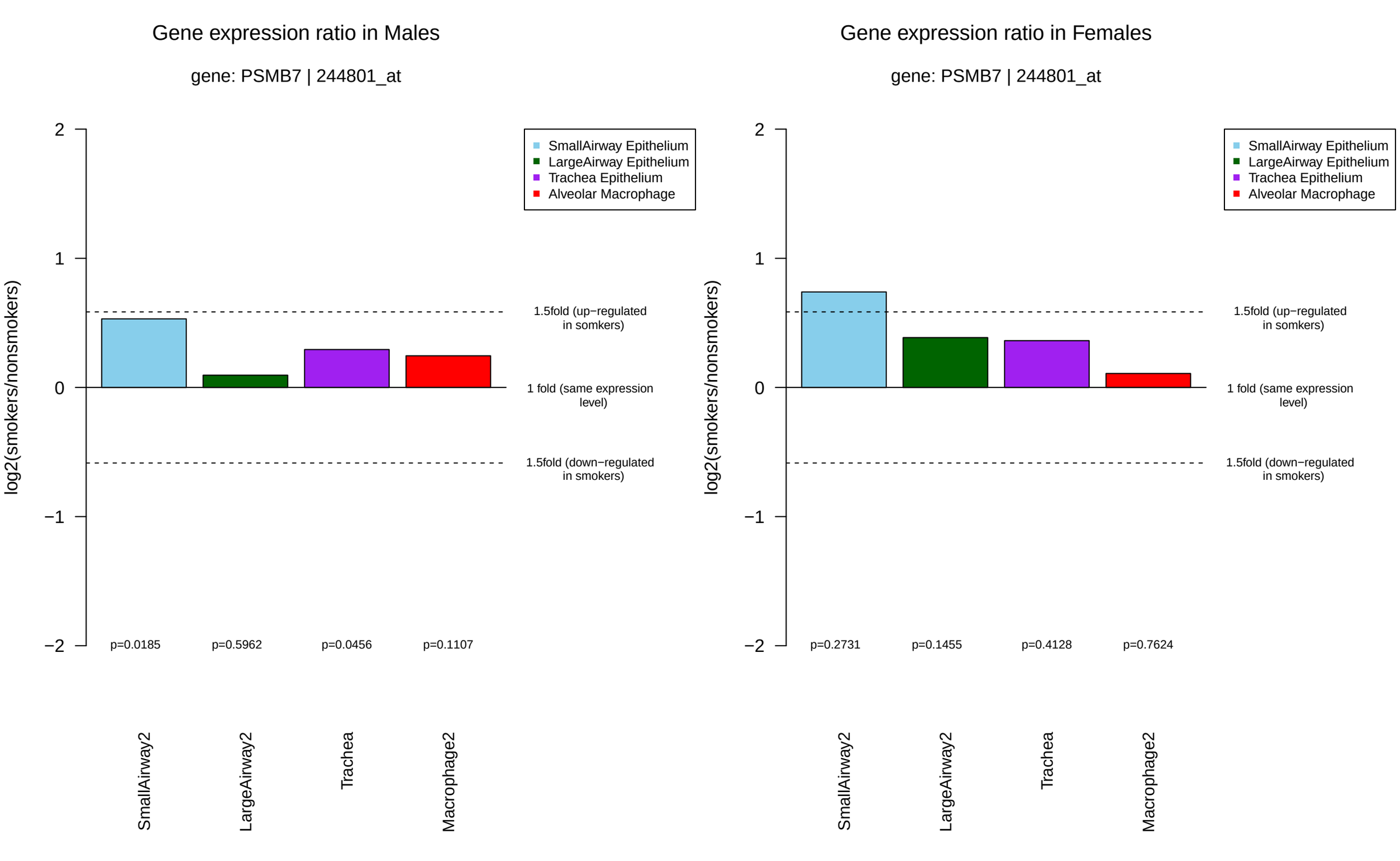
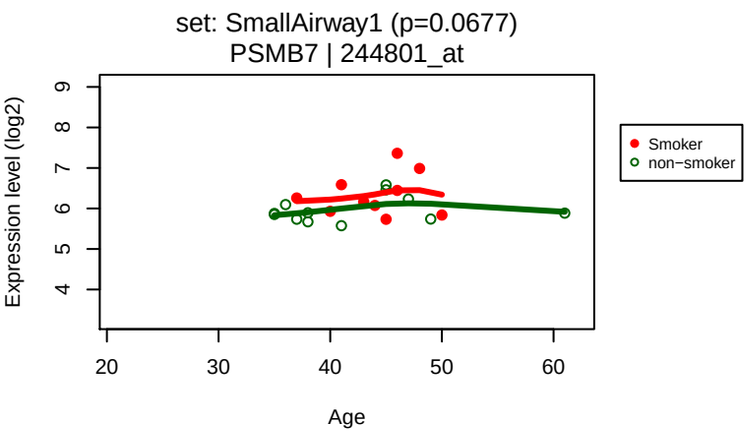
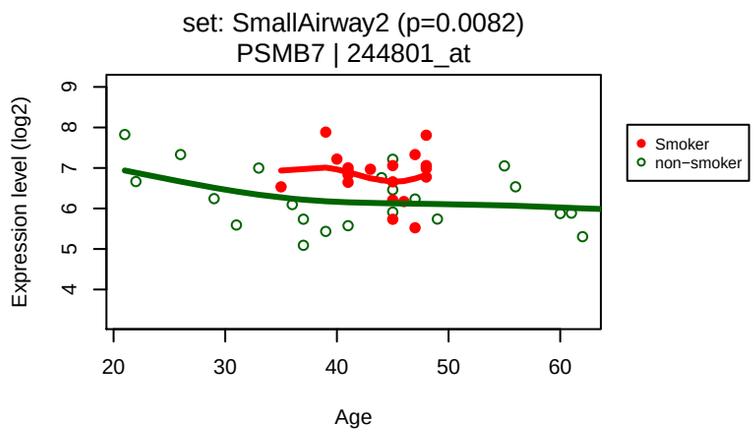
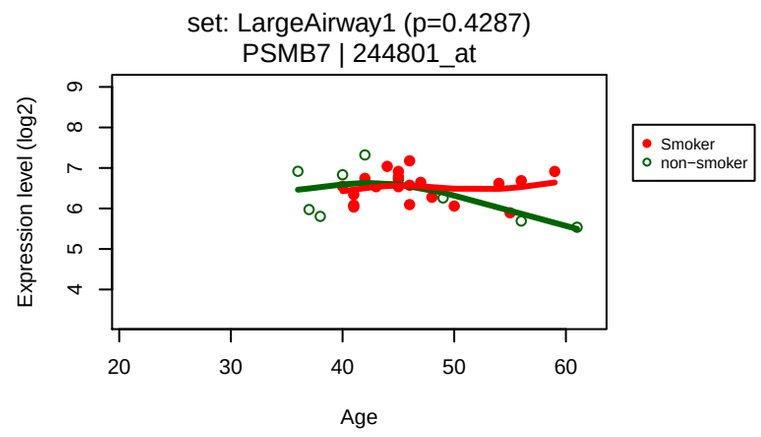

Gene expression of smokers and nonsmokers
gene: PSMB7 | 244801_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0677
p=0.0082
p=0.0149
p=0.4287
p=0.2679
p=0.0125
p=0.0027
p=0.1542
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

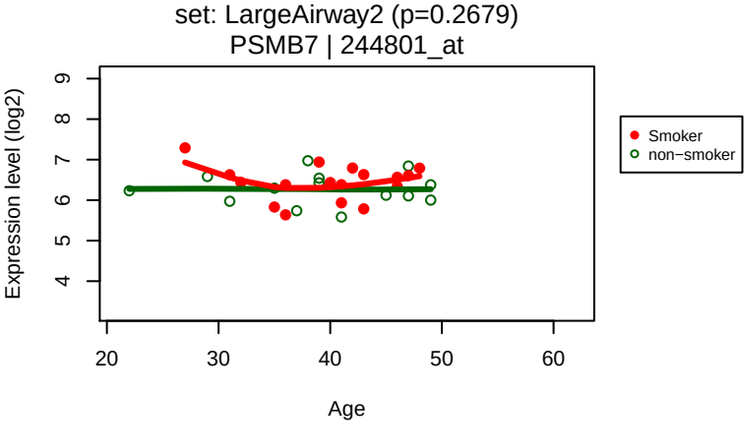

Gene expression ratio (smokers/nonsmokers)
gene: PSMB7 | 244801_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0677
p=0.0082
p=0.0149
p=0.4287
p=0.2679
p=0.0125
p=0.0027
p=0.1542
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

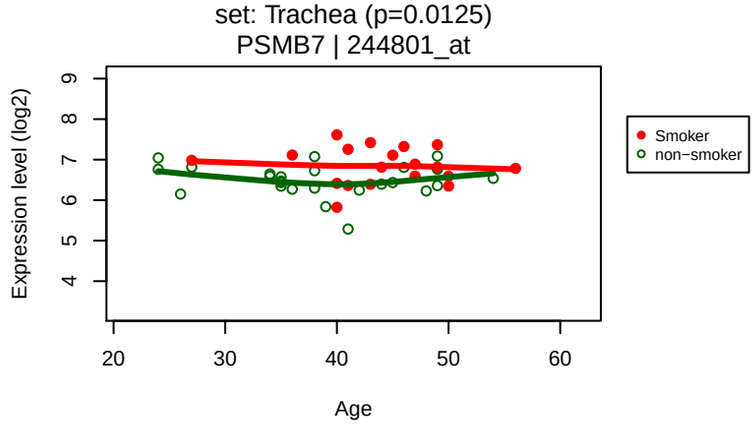

Gene expression ratio in Males
gene: PSMB7 | 244801_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0185
p=0.5962
p=0.0456
p=0.1107
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB7 | 244801_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2731
p=0.1455
p=0.4128
p=0.7624
SmallAirway2
LargeAirway2
Trachea
Macrophage2

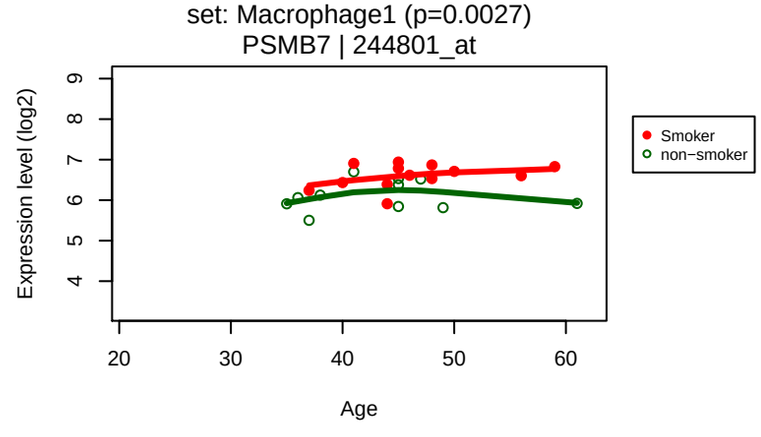

set: SmallAirway1 (p=0.0677)
PSMB7 | 244801_at
Expression level (log2)
Age
Smoker
non−smoker

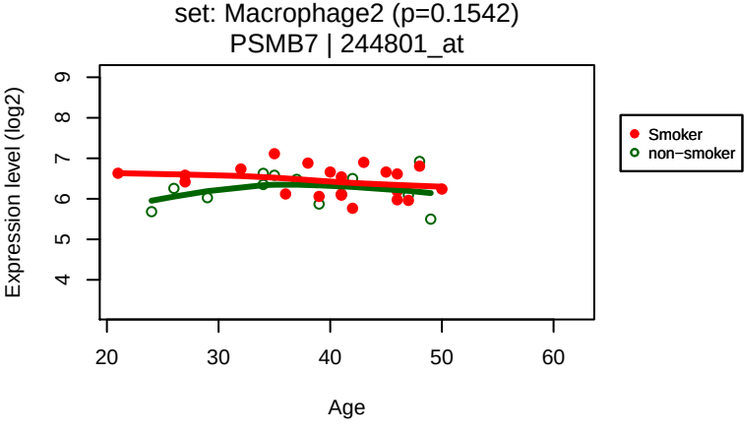

set: SmallAirway2 (p=0.0082)
PSMB7 | 244801_at
Expression level (log2)
Age
Smoker
non−smoker

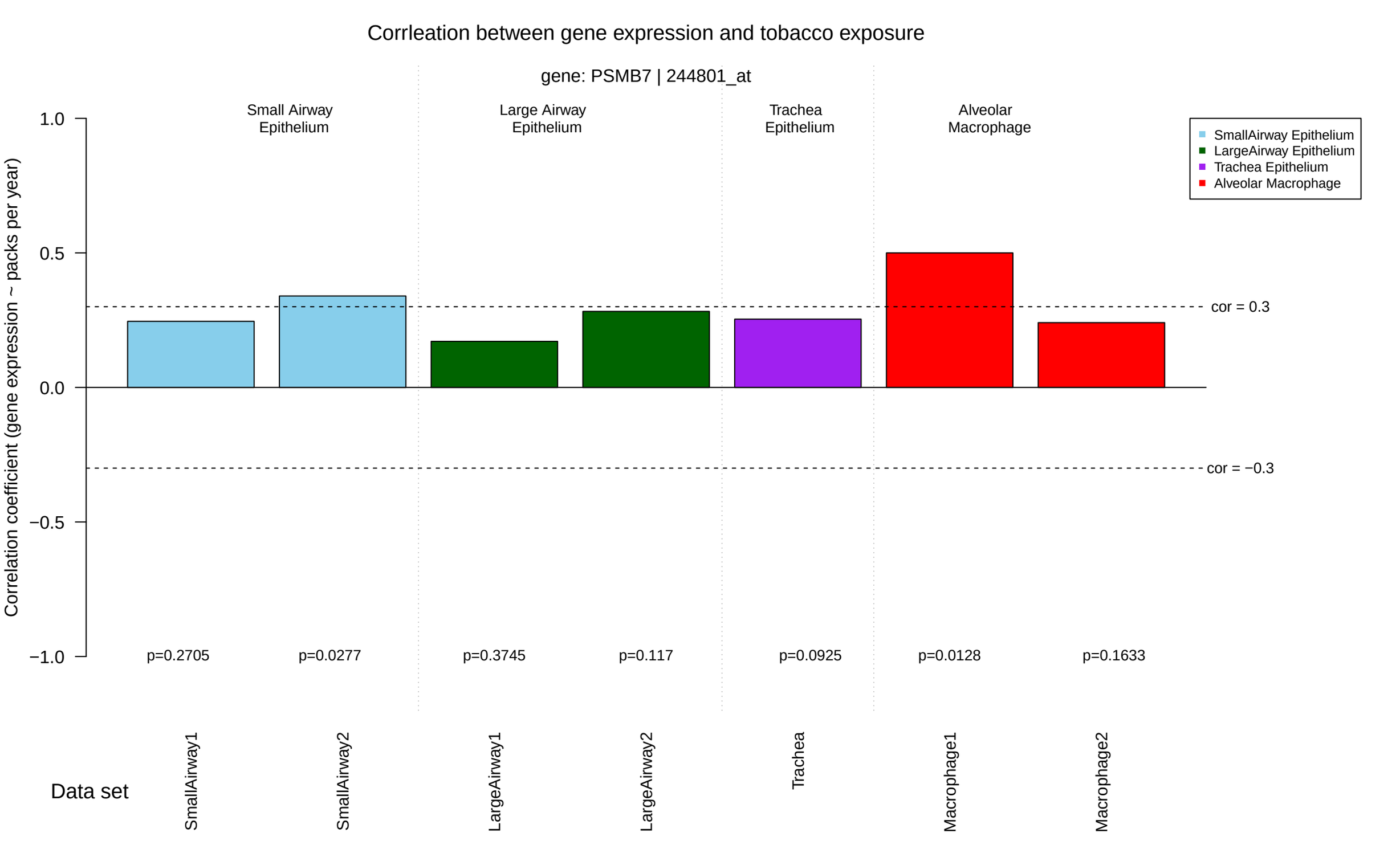

set: LargeAirway1 (p=0.4287)
PSMB7 | 244801_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2679)
PSMB7 | 244801_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0125)
PSMB7 | 244801_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0027)
PSMB7 | 244801_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1542)
PSMB7 | 244801_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB7 | 244801_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2705
p=0.0277
p=0.3745
p=0.117
p=0.0925
p=0.0128
p=0.1633
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2